

Inter-Subject Correlation of Heart Rate and Attention Modulation

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Abstract

This project focuses on analyzing heart-rate synchrony across individuals to verify if it's modulated by attentional state while watching educational videos.

The goal is therefore to reproduce a couple of results from Pérez et al. (2021) [1] using data from the Brain, Body and Behavior Dataset (BBBD) [2], which provides a standardized and publicly accessible version of the recordings used in the original Experiment 2.

To achieve this, heart-rate fluctuations were extracted from raw ECG recordings and compared across subjects using ISC-HR (inter-subject correlation of heart-rate).

A hypothesis test with a FDR correction is used to establish statistical significance of the obtained results. ISC-HR is then compared between attentive and distracted conditions using paired statistical tests for each of the five stimuli.

Hypothesis test and descriptive statistics are employed to analyze synchrony patterns and to detect the effect of attention across stimuli.

The results provide a statistical reproduction of the ISC-HR modulation by attention reported by Pérez et al. (2021).

1 Introduction

Heart rate is often treated as a purely autonomic signal, reflecting internal physiological states such as arousal or stress.

A recent study suggests that heart-rate fluctuations can be driven by external stimuli. When multiple individuals are exposed to the same narrative content, their heart rates tend to fluctuate in a similar way.

This phenomenon is known as inter-subject correlation of heart-rate (ISC-HR).

Pérez et al. (2021) [1] proposed that these heart-rate fluctuations are modulated by conscious, attentive processing of the stimulus.

This project aims to reproduce two of the main results reported by the analyzed paper:

- ISC-HR is statistically significant across subjects exposed to the same stimulus (I'll define it as R1).
- Synchrony is modulated by attentional state, being higher when subjects are attentive rather than when they are distracted (I'll define it as R2).

This work uses data from the Brain, Body and Behavior Dataset (BBBD) [2], which is a more recent dataset that includes similar experiments to the original study.

2 Related Work

The original paper [1] computes ISC-HR by correlating each subject's instantaneous heart-rate fluctuations against the rest of the group, averaging the resulting pairwise correlations in Fisher-Z space, and testing significance via a circular-shift permutation procedure with False Discovery Rate (FDR) correction. The study reports three main findings across four experiments.

The first one (R1) shows that heart-rate fluctuations synchronize across subjects listening to the same auditory narrative, with 17 of 27 subjects having statistically significant ISC-HR.

The second result (R2) shows this synchrony is systematically higher when subjects attend to a stimulus rather than when they are distracted.

The third result (R3) shows that ISC-HR predicts subsequent memory recall performance for the narrative content, suggesting that the measure reflects conscious engagement rather than an autonomic response to sensory input.

The fourth experiment applies ISC-HR to patients with disorders of consciousness, showing reduced synchrony relative to healthy controls and framing ISC-HR as a potential prognostic marker in a clinical setting.

As already anticipated, this project reproduces only R1 and R2 using an experiment equivalent to the original study's experiment 2.

3 Dataset

This project uses Experiment 2 of the BBBD dataset [2], which contains recordings of ECG, EEG, eye-tracking, and respiration during educational video viewing.

In experiment 2, 31 subjects watch five educational videos, each one viewed twice by the same subject: attentively the first time, distracted by performing a secondary task (silently counting backward) the second time.

ECG recordings were sampled at 128 Hz, with stimulus duration ranging from approximately 2.5 to 6.5 minutes depending on the video.

BBBD’s data descriptor [2] explicitly states that the ECG data provided for this experiment are not newly collected, but are the same recordings originally published in Pérez et al., 2021 [1].

ECG preprocessing documentation [2] states that R-wave peaks were detected automatically and then manually corrected. This project instead detects R-peaks directly from the raw ECG signal using an automated procedure, without using BBBD’s provided peak timestamps and without manual review.

This choice was made in order to reproduce and understand the original study’s methodology.

Not every subject has a usable recording for every combination (subject, session, stimulus). Missing files are skipped automatically, so the number of subjects analyzed varies slightly across stimuli and conditions.

4 Methodology

This chapter describes the full process from raw ECG signals to the statistical tests used to reproduce R1 and R2.

Most of the process follows the methodology described in the original paper.

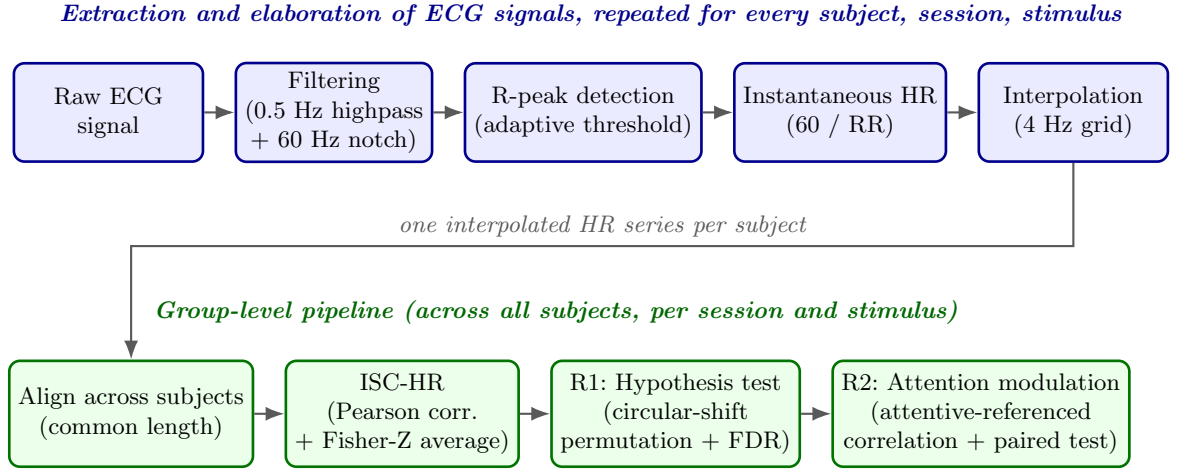


Fig. 1 Full processing pipeline from raw ECG signal to the statistical tests to reproduce R1 and R2.

4.1 Signal preprocessing

Raw ECG signals are first high-pass filtered to remove baseline drift, then a notch filter is used to suppress power-line interference. The first high-pass filter used a 0.5 Hz cutoff, in order to preserve the faster fluctuations corresponding to heartbeats while attenuating baseline drift. The notch filter targets 60 Hz, which is the frequency of electrical interference from the power supply. Its contribution is expected to be limited because the interference frequency is very close to Nyquist limit.

R-peaks are detected by using an adaptive amplitude threshold. The chosen value is $3 \times$ standard deviations of the filtered signal. This is combined with a 0.3s temporal distance between consecutive detections.

Instantaneous heart rate was computed as the inverse of consecutive R-R intervals $\frac{60}{RR}$ (BPM), where RR is the time interval between two consecutive R-peaks. Then it's resampled onto a common 4Hz grid via interpolation to allow comparison across subjects.

This last step is made because instantaneous HR series have one value per heartbeat and themselves do not occur at fixed intervals. The requirement for the next step which is computing a Pearson correlation between the HR signals of different subjects requires vectors of equal length, aligned on the same time points.

Subjects whose mean heart rate fell outside a plausible range (40-160 BPM) were automatically excluded from further analysis in order to improve the quality of the results.

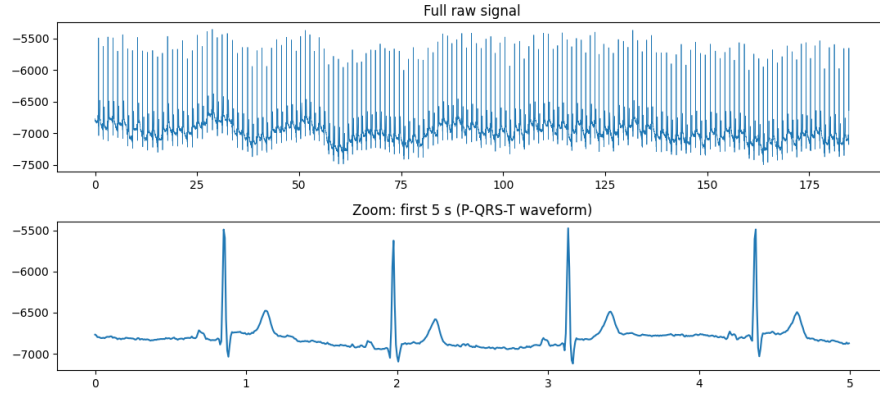


Fig. 2 Full raw ECG signal of a single subject on a single session and one stimulus.

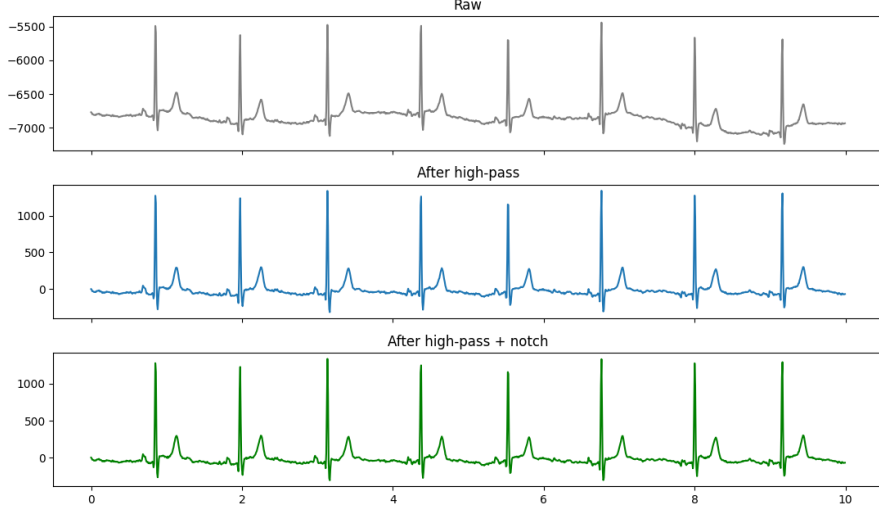


Fig. 3 Raw ECG signal and after filters are applied.

4.2 ISC-HR computation

In order to compute ISC-HR, all subjects' interpolated HR series are truncated to their shortest common length and arranged into a matrix of pairwise Pearson correlations. Each subject's ISC-HR is computed as follows:

$$\text{ISC-HR}_i = \tanh \left(\frac{1}{N-1} \sum_{j \neq i} \text{arctanh}(r_{ij}) \right) \quad (1)$$

where r_{ij} is the Pearson correlation between subject i and subject j and self-correlation ($j = i$) is excluded. Correlations are first transformed with the Fisher-Z transform (arctanh) before averaging. The inverse transform ($\tanh$) then maps the average back onto the original correlation scale.

4.3 Hypothesis test

In order to establish statistical significance of each subject's ISC-HR, a circular-shift permutation test is computed.

The subject's HR series is circularly shifted by a random amount and re-correlated against the reference group. This is repeated $B = 10,000$ times to build a null distribution, which is the set of ISC-HR values that subject would obtain purely by chance.

The p-value is the fraction of the B null values that are at least as large as the subject's real observed ISC-HR. It measures how surprising the observed value is under the assumption of no true synchrony.

A small p-value means it would be rare, by chance alone, to obtain a value this large. In practice, this fraction is computed as:

$$p = \frac{1 + \text{count}}{B + 1} \quad (2)$$

where count is the number of null values at least as large as the observed ISC-HR. If the observed value exceeds all B permutations, the plain proportion returns exactly $p = 0$, which incorrectly implies that chance could never produce such a result. The $+1$ correction bounds the smallest reportable p-value at $\frac{1}{B+1}$ instead. Benjamini-Hochberg FDR correction is then applied across all subjects within each combination.

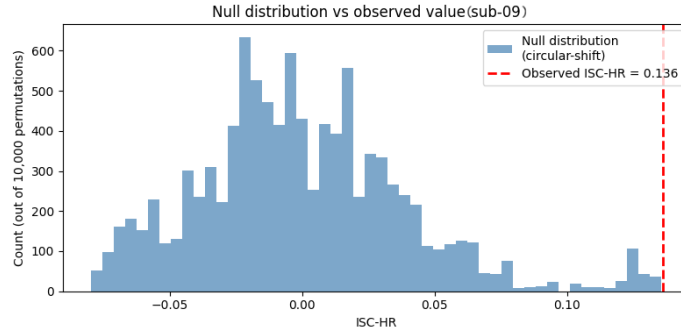


Fig. 4 Example of null distribution generated showing the subject with the highest ISC-HR value. The observed value lies in the extreme tail of the null distribution, indicating that it is unlikely under the null hypothesis.

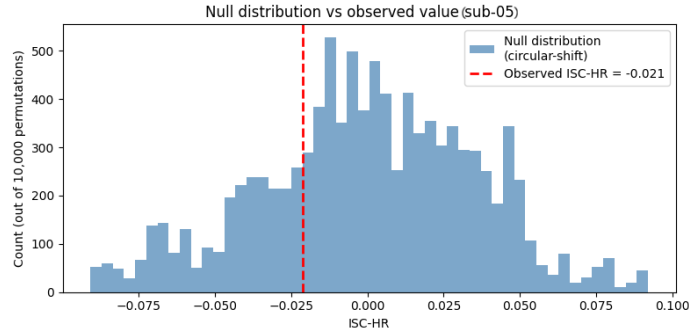


Fig. 5 Another example of null distribution which say the opposite. Since the value stands right in the middle of the distribution, we can assume it's not significant.

4.4 Attention modulation

Following Pérez et al., 2021 [1] STAR Methods, ISC-HR for the distracted condition was computed against a fixed reference: each distracted subject’s series was correlated against the entire attentive group, rather than against other distracted subjects.

For each of the five stimuli, the attentive-minus-distracted ISC-HR difference was computed per subject (restricted to subjects with a usable recording in both conditions). A Shapiro-Wilk normality test was run once on the pooled differences across all five stimuli to decide between a paired t -test and a Wilcoxon signed-rank test. The chosen test is then applied uniformly to each stimulus individually, testing whether the difference between the attentive and distracted ISC-HR is significantly different from zero.

A significant result ($p < 0.05$) for a given stimulus indicates that attention modulates ISC-HR for that stimulus, addressing R2.

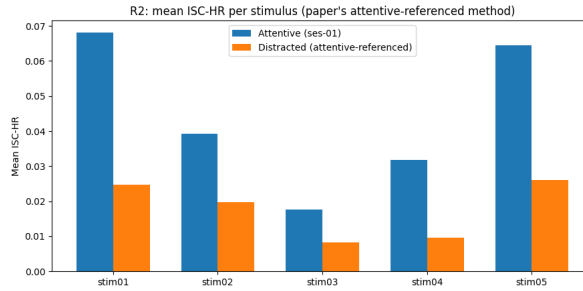


Fig. 6 Mean ISC-HR per stimulus in the attentive condition and in the distracted condition. Each distracted subject is correlated against the fixed attentive group instead of against other distracted subjects.

In addition to the paired tests, an ANOVA (attention $\times$ stimulus) is used on the same ISC-HR values, following the same fixed-effects structure reported by Pérez et al., 2021 [1].

Since ANOVA requires a fully balanced design, only subjects with a complete set of observations across all five stimuli and both conditions are retained.

4.5 Implementation choices

Several parameters were not specified in the original paper and required to make implementation choices:

- The amplitude threshold and refractory distance for R-peak detection.
- Physiological HR range used for quality control.

Two other choices were also made for simplicity:

- Regarding R1, ISC-HR in the distracted condition is computed within-group (distracted subjects correlated against each other) rather than using the attentive-referenced method. The latter is used specifically to reproduce R2, where the attentive-distracted comparison requires it.
- This project’s ANOVA reproduces the paper’s fixed effects (attention, stimulus) and additive structure, but not its random subject effect.

5 Results

This section reports the results of R1 (Section 5.1) and R2 (Section 5.2) across all ten (session, stimulus) combinations.

5.1 R1 on all combinations of session and stimulus

Figure 7 shows an example of correlation matrix, computed across all subjects for the attentive condition (session 1) and stimulus 1.

ISC-HR for each subject is derived from this matrix as described in Section 4.2.

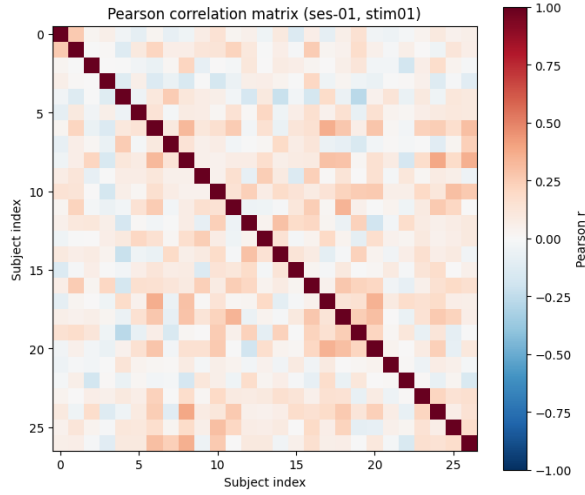


Fig. 7 Pearson correlation matrix for session 1 (attentive), stimulus 1. Entry (i, j) is the Pearson correlation between subject i and subject j . The dark diagonal is each subject with itself ($r = 1$).

Table 1 summarizes ISC-HR results across all five stimuli, for both attentive and distracted conditions. In this specific case, as already anticipated in the previous sections, the distracted condition is correlated within-group, not against the attentive reference.

For every stimulus, the attentive condition shows a positive mean ISC-HR with a substantial fraction of individually significant subjects (from 17.9 to 48.1%), well above the $\sim 5\%$ expected by chance alone at $\alpha = 0.05$. The distracted condition, computed within-group, shows essentially no synchrony for any stimulus.

Stimulus	Condition	Subjects	Mean ISC-HR	Significant (%)
stim01	Attentive	27	0.068	48.1
stim01	Distracted	25	0.002	0.0
stim02	Attentive	27	0.039	18.5
stim02	Distracted	27	-0.002	0.0
stim03	Attentive	28	0.018	17.9
stim03	Distracted	25	0.003	0.0
stim04	Attentive	28	0.032	25.0
stim04	Distracted	25	0.002	0.0
stim05	Attentive	28	0.064	21.4
stim05	Distracted	26	0.008	0.0

Table 1 ISC-HR results across all ten combinations (within-group method for both conditions).

5.2 R2: attention modulation

Table 2 shows, for each stimulus, the mean ISC-HR under both the attentive condition and the attentive-referenced distracted condition, together with the result of the paired comparison between them.

Stimulus	Attentive	Att. sig. (%)	Distracted (ref.)	Dist. sig. (%)	<i>p</i> -value
stim01	0.068	48.1	0.025	4.0	0.0004
stim02	0.039	22.2	0.020	22.2	0.2584
stim03	0.018	17.9	0.008	0.0	0.0802
stim04	0.032	25.0	0.010	0.0	0.0003
stim05	0.064	21.4	0.026	0.0	0.0236

Table 2 Mean ISC-HR and percentage of individually significant subjects (Section 4.3) for the attentive and attentive-referenced distracted conditions, alongside the paired comparison between them, per stimulus. The Wilcoxon signed-rank test is used, chosen over a paired *t*-test after Shapiro-Wilk rejected normality on the pooled differences at $\alpha = 0.05$ ($W = 0.979$, $p = 0.046$; W close to 1 indicates normality, so this result sits just below the threshold).

Three of the five stimuli (stim01, stim04, stim05) show a statistically significant reduction in ISC-HR under distraction ($p < 0.05$), consistent with R2. The remaining two stimuli do not reach significance, for different reasons discussed below.

The ANOVA results are reported below.

- Attention ($F(1, 23) = 20.18$, $p < 0.001$)
- Stimulus ($F(4, 92) = 4.27$, $p = 0.003$)
- Attention $\times$ Stimulus ($F(4, 92) = 1.04$, $p = 0.392$)

Both main effects (Attention and Stimulus) are significant. The interaction term is not significant.

Only 24 subjects in R2 have a complete design across all five stimuli and both conditions, while the remaining 4 are dropped because a fully balanced design is required.

5.2.1 stim02 and stim03: possible reasons for the lack of significance

Stim02 and stim03 both fail to reach significance ($p = 0.258$ and $p = 0.080$ respectively), but for different underlying reasons.

For stim02, the attentive and distracted conditions did not differ significantly, although the absence of a significant effect does not by itself demonstrate preserved attentional engagement.

One possible explanation of stim03’s weaker result is the lower attentive baseline (0.018, the smallest of all five stimuli), which limits how large the attentive/distracted difference can be.

6 Discussion

6.1 Comparison with the original study

Attentive viewing produces significant inter-subject heart-rate synchrony which is reduced under distraction. However, direct numerical comparison with the paper’s reported figures should be treated with some caution. BBBD’s data descriptor explicitly attributes Experiment 2’s ECG data to Pérez et al., 2021 [1] (Section 3), confirming the same underlying raw recordings used here.

However, the paper we based the work on, does not specify the exact peak detection parameters (amplitude threshold, refractory distance) used.

Exact numerical agreement with the original study is therefore not guaranteed, even on the same raw signal, since this project’s peak-detection parameters (Section 4.5) are an independent choice.

The original paper reports that the effect of attention is significant for each video individually (pairwise t -test, all $p < 0.05$), and that ISC-HR numerically drops under distraction for 26 of 27 subjects when averaged across videos.

In this project instead the paired comparison reached significance for 3 of 5 stimuli.

Regarding the significance of the ISC-HR synchrony across subjects, we compare against experiment 1 of the paper, which is an audio only narrative without distraction session. The paper reports 17 of 27 subjects (63%) having statistically significant HR correlation, which is comparable with what this project found in attentive condition (from 17.9% to 48.1% depending on stimulus, Table 1). However, it is clearly not a direct like for like comparison.

The ANOVA’s main effects agree with the original paper in direction and significance. As already mentioned in Section 5.2 both main effects are confirmed. The reported F-statistics are not directly comparable in magnitude, since the paper’s model includes a random subject effect not replicated here.

Attention significantly affects ISC-HR and so does the stimulus. This, united with per-stimulus paired tests, shows that the attention effect holds not only stimulus by stimulus, but also when all five stimuli are analyzed jointly in a single model.

The interaction between attention and stimulus is not significant, meaning the size of the attention effect may not depend on which video is being watched.

The differences from the original study’s R1 and R2 figures noted above are plausibly

attributable to the peak-detection parameter approaches (Section 4.1) and other criteria already mentioned in previous sections.

In addition, the BBBBD data descriptor was published several years after the original paper, although the Experiment 2 recordings themselves originate from the original study. The timing difference is a further reason to treat numerical comparison as plausibility check rather than exact validation.

6.2 A methodological note on Tables 1 and 2

Stim02’s attentive-condition significance rate differs slightly between Table 1 (18.5%) and Table 2 (22.2%), despite both being computed from the same standard within-group correlation. This is because the two tables align series against different companion groups: Table 1 aligns attentive subjects alone, while Table 2 aligns them together with the distracted group, as required for the cross-condition correlation used in R2. A shorter common length in one case versus the other can shift a single borderline subject’s significance verdict. This affects only stim02, by one subject, and does not change any conclusion drawn from either table.

7 Conclusion

At the end of this project, we can assume that the results are consistent with the hypothesis that heart-rate synchrony is modulated by attentional engagement.

Positive ISC-HR is observed in the attentive condition for all five stimuli, whereas the distracted within-group condition shows values close to zero. This gap is statistically confirmed for the majority of the stimuli in R2.

Regarding the exceptions found, stim02 shows no significant difference between conditions, which does not by itself show that engagement persisted. The pattern observed here is consistent with attention playing a role in heart-rate synchronization.

Overall, this project reproduces the qualitative pattern reported by Pérez et al., 2021 [1].

Heart rate synchrony is measurable, statistically significant and reduced under distraction. The magnitude of the effect differs somewhat from the original figures, as already discussed, plausibly due to differing implementation choices such as peak-detection parameters and ordinary sample variability (Section 6.1). However the direction and general strength of both R1 and R2 are consistent with the original study.

References

- [1] P. Pérez, J. Madsen, L. Banellis, B. Türker, F. Raimondo, V. Perlberg, M. Valente, M.-C. Niérat, L. Puybasset, L. Naccache, T. Similowski, D. Cruse, L. C. Parra, and J. D. Sitt. Conscious processing of narrative stimuli synchronizes heart rate between individuals. *Cell Reports*, 36(11):109692, 2021.
- [2] J. Madsen, N. Kuppa, and L. C. Parra. The Brain, Body, and Behavior Dataset (BBBD): Multimodal Recordings during Educational Videos. *Scientific Data*, 13:920, 2026.